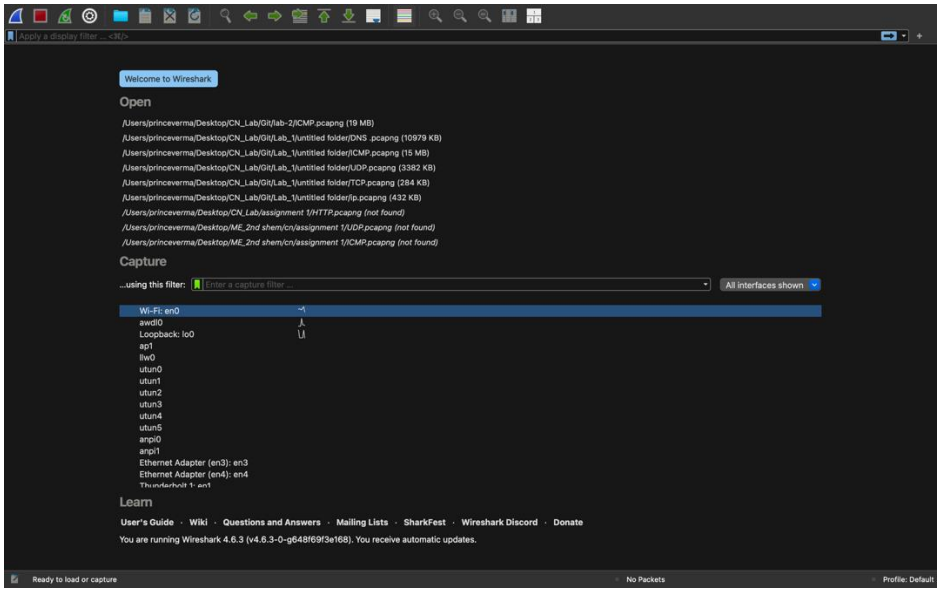
CN ASSIGNMENT 1

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

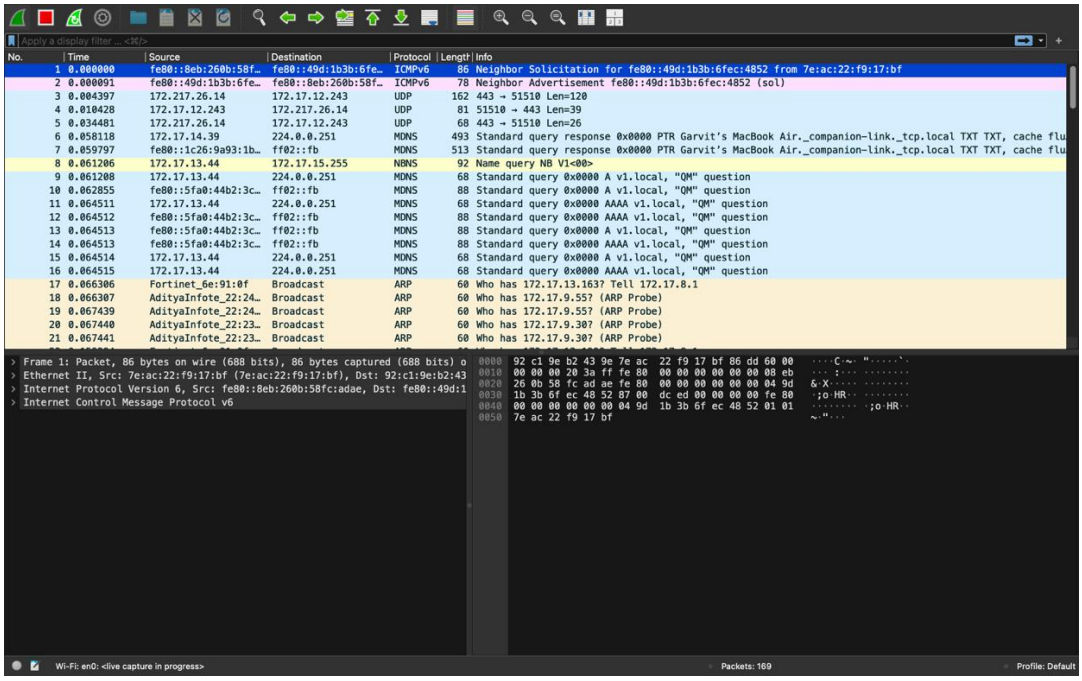
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
    - Wi-Fi
    - Ethernet



Step 2: Start Packet Capture

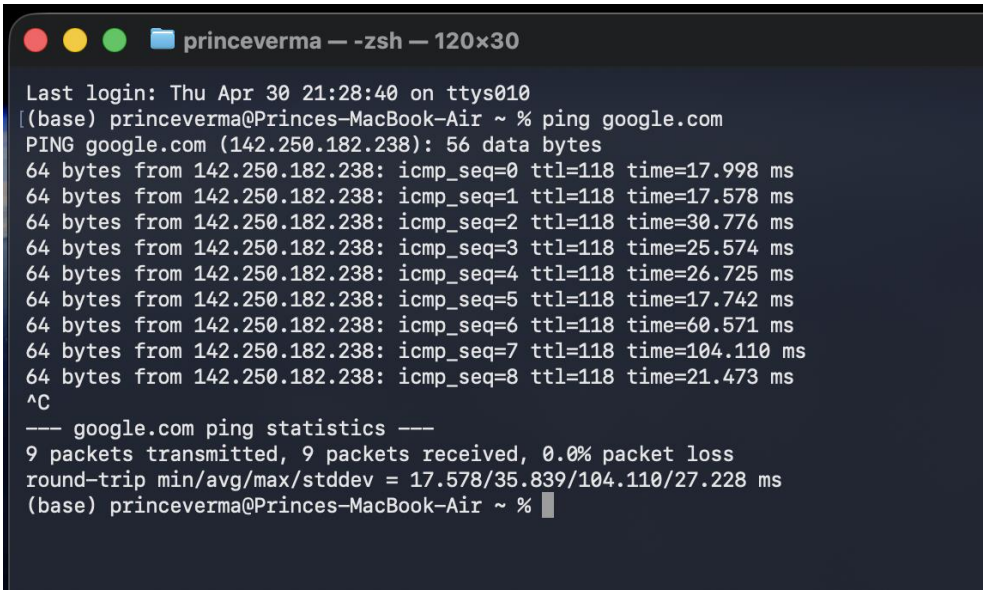
- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

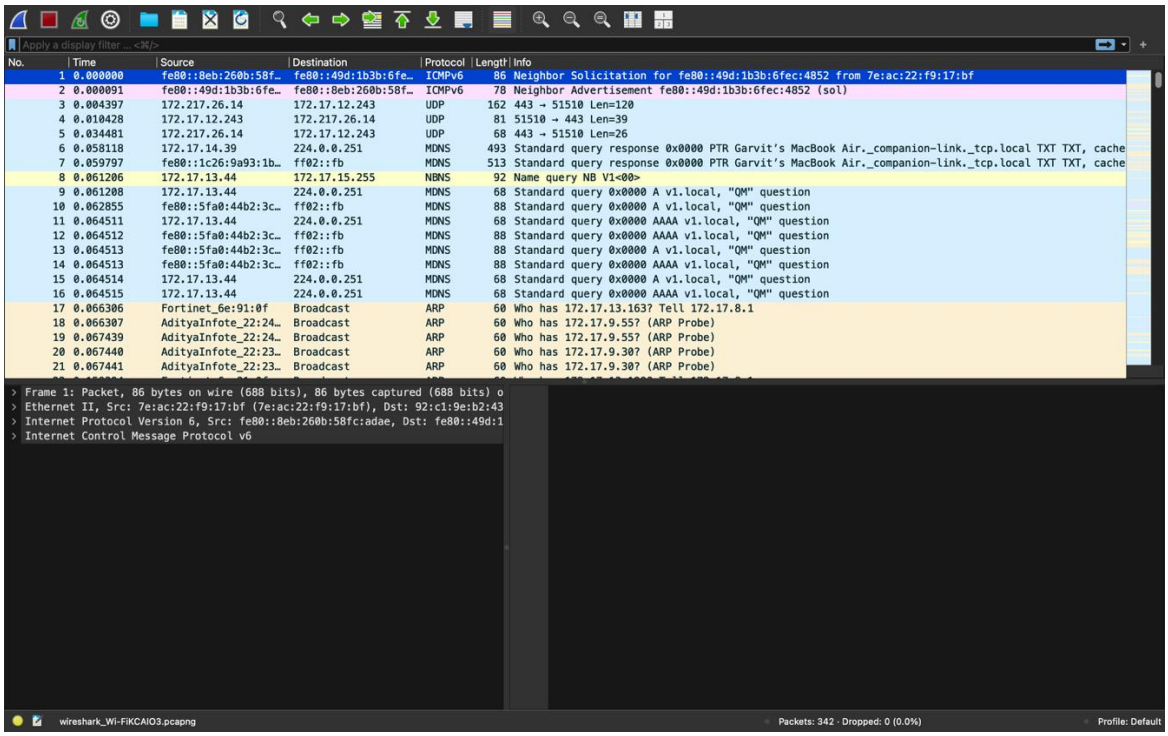
Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com



Step 4: Stop Capture

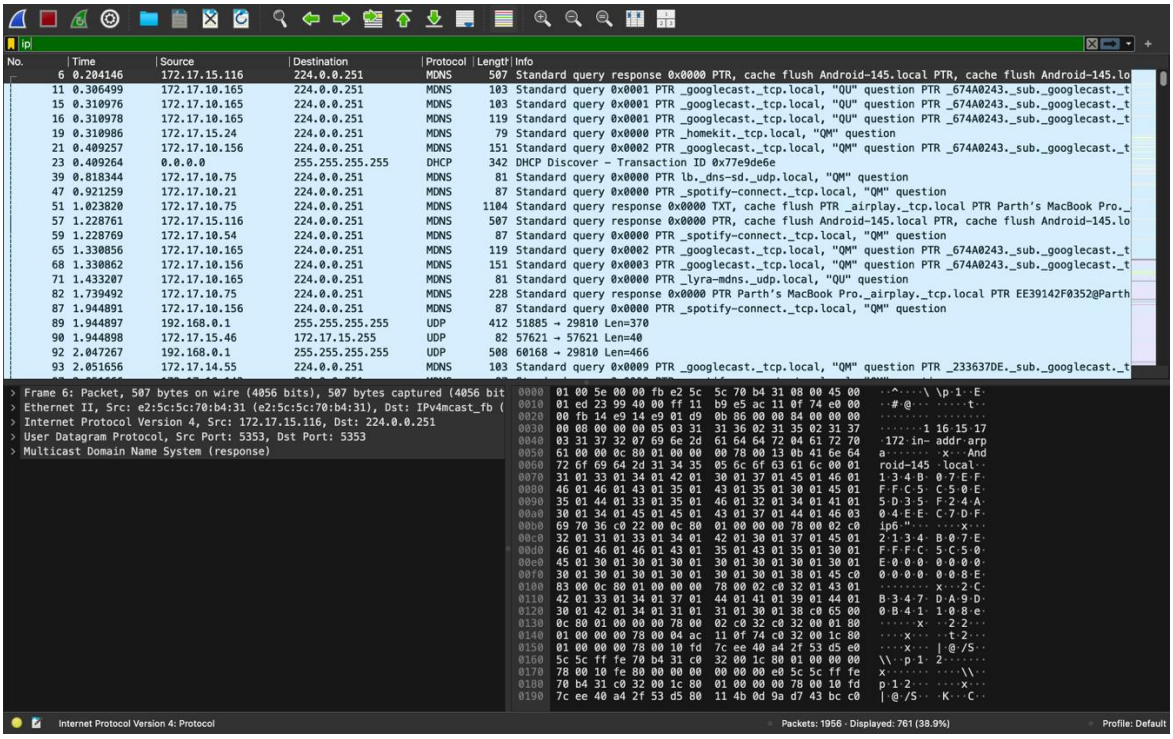
- Click the **red Stop button**
- Captured packets will remain on the screen



Step 5: Apply Basic Filters

Use the **Display Filter bar**:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets



tcp

No.	Time	Source	Destination	Protocol	Length	Info
333	3.179590	57.144.147.33	172.17.10.47	TCP	912	5222 → 63712 [PSH, ACK] Seq=1 Ack=1 Win=786 Len=846 TSval=1005774844 TSecr=4154813548 [TCP PDU re
334	3.179883	172.17.10.47	57.144.147.33	TCP	66	63712 → 5222 [ACK] Seq=1 Ack=847 Win=2035 Len=0 TSval=4154880801 TSecr=1005774844
342	3.512625	172.17.10.47	57.144.147.33	TCP	129	63712 → 5222 [PSH, ACK] Seq=1 Ack=847 Win=2048 Len=63 TSval=4154881134 TSecr=1005774844
343	3.534778	57.144.147.33	172.17.10.47	TCP	66	5222 → 63712 [ACK] Seq=847 Ack=64 Win=786 Len=0 TSval=1005775231 TSecr=4154881134
512	7.886489	52.168.112.66	172.17.10.47	TCP	66	443 → 63865 [FIN, ACK] Seq=1 Ack=1 Win=102 Len=0 TSval=652601722 TSecr=869984244
513	7.886841	172.17.10.47	52.168.112.66	TCP	66	63865 → 443 [ACK] Seq=1 Ack=2 Win=2048 Len=0 TSval=870112461 TSecr=652601722
514	7.887702	172.17.10.47	52.168.112.66	TLSv1…	90	Application Data
515	7.888411	172.17.10.47	52.168.112.66	TCP	66	63865 → 443 [FIN, ACK] Seq=25 Ack=2 Win=2048 Len=0 TSval=870112463 TSecr=652601722
516	7.890463	52.168.112.66	172.17.10.47	TCP	66	443 → 63865 [ACK] Seq=2 Ack=25 Win=102 Len=0 TSval=652601733 TSecr=870112462
517	7.892132	52.168.112.66	172.17.10.47	TCP	66	443 → 63865 [ACK] Seq=2 Ack=26 Win=102 Len=0 TSval=652601733 TSecr=870112463
894	13.583950	172.17.10.47	57.144.147.32	TCP	78	63876 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3107425095 TSecr=0 SACK_PE
895	13.587693	57.144.147.32	172.17.10.47	TCP	74	443 → 63876 [SYN, ACK, ECE] Seq=0 Ack=1 Win=14480 Len=0 MSS=1460 SACK_PERM TSval=652602302 TSecr=
896	13.588669	172.17.10.47	57.144.147.32	TCP	66	63876 → 443 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=3107425099 TSecr=652602302
897	13.588709	172.17.10.47	57.144.147.32	TCP	1514	63876 → 443 [ACK] Seq=1 Ack=1 Win=131776 Len=1448 TSval=3107425100 TSecr=652602302 [TCP PDU reass
898	13.588715	172.17.10.47	57.144.147.32	TLSv1…	159	Client Hello (SNI=graph.whatsapp.net)
899	13.592679	57.144.147.32	172.17.10.47	TCP	66	443 → 63876 [ACK] Seq=1 Ack=1449 Win=17408 Len=0 TSval=652602303 TSecr=3107425100
900	13.592682	57.144.147.32	172.17.10.47	TCP	66	443 → 63876 [ACK] Seq=1 Ack=1542 Win=17408 Len=0 TSval=652602303 TSecr=3107425100
909	13.644990	57.144.147.32	172.17.10.47	TLSv1…	1514	Server Hello, Change Cipher Spec
910	13.644994	57.144.147.32	172.17.10.47	TLSv1…	1378	Application Data
911	13.644997	57.144.147.32	172.17.10.47	TLSv1…	811	Application Data
912	13.646454	172.17.10.47	57.144.147.32	TCP	66	63876 → 443 [ACK] Seq=1542 Ack=3506 Win=128320 Len=0 TSval=3107425157 TSecr=652602306

```
> Frame 333: Packet, 912 bytes on wire (7296 bits), 912 bytes captured (7296 b
> Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: 0e:35:09:0c:20
> Internet Protocol Version 4, Src: 57.144.147.33, Dst: 172.17.10.47
> Transmission Control Protocol, Src Port: 5222, Dst Port: 63712, Seq: 1, Ack:
```

```
0000  0e 35 09 0c 20 e2 04 d5  90 6e 91 0f 08 00 45 02   ·5·· ··· ·n····E·
0010  03 82 43 dc 40 00 38 06  78 a6 39 90 93 21 ac 11   ··C·@·8· x·9··!··
0020  0a 2f 14 66 f8 e0 e4 0d  62 7b be 71 67 d6 80 18   ·/·f···· b{·qg···
0030  03 12 46 7b 00 00 01 01  08 0a 3b f2 e7 fc f7 a5   ··F{···· ··;·····
0040  6c 6c 00 03 4b 70 a8 33  d6 94 a2 d9 b5 53 c9 9d   ll··Kp·3 ·····S··
0050  dc d9 54 ec 2c 94 de eb  ec fb f6 1c 80 83 a2 15   ··T·,··· ........
0060  61 5c d3 bd 3f 27 9b 00  84 f2 b7 cf 4b 3b 63 05   a\··?'·· ····K;c·
0070  e3 8d cb 55 45 ae e4 97  16 8c d2 84 71 30 bd af   ···UE··· ····q0··
0080  23 28 f7 56 a5 89 ef da  fb 32 4d d5 f9 68 80 88   #(·V···· ·2M··h··
0090  b9 73 c1 15 16 7e f4 db  a3 5e 76 8f 4f 42 bf 23   ·s···~·· ^v·OB·#
00a0  cf 1e 3f cb b4 4d fe 65  f4 0b 3e c0 6b 17 52 9d   ··?··M·e ··>·k·R·
00b0  d4 d1 51 98 c6 40 15 df  24 70 24 c2 13 6a ef c3   ··Q··@·· $p$··j··
00c0  e4 d4 df b5 7d 36 0b c7  99 65 0f 63 1b bd 60 42   ····}6·· ·e·c··`B
00d0  79 8a d0 26 96 07 6c 1d  b3 38 9b b9 43 1c 8e f8   y··&··l· ·8··C···
00e0  97 79 29 89 ed 5a 67 19  96 16 8e cf d9 0d 7f 83   ·y)··Zg· ........
00f0  92 c9 9b 19 0e 36 34 c9  10 03 59 c2 92 9d a5 7d   ·····64· ··Y···}
0100  64 6f eb 87 9c 9a ea 1d  80 96 21 35 16 89 d5 24   do····· ··!5···$
0110  95 70 0d 6e 97 d3 87 b9  6b 97 73 82 66 3c 35 a5   ·p·n···· k·s·f<5·
0120  67 42 f4 44 8f 10 58 73  32 0a eb 19 b6 d9 c9 e4   gB·D··Xs 2·····
0130  ec 7d 9c 90 97 df fd b9  9f 00 4a 45 68 8d 96 56   ·}····· ··JEh··V
0140  7a 7e 0a cb 5c 18 ae 13  85 e3 12 7e 7d 57 6b b0   z~··\··· ···~}Wk·
0150  cf a5 4d 0d 61 85 6a b1  32 9c 61 ce 66 f1 09 ca   ··M·a·j· 2·a·f···
0160  a5 f8 07 1a fe e7 f9 af  35 af 62 a8 df ef 3f 19   ····· 5·b···?·
0170  33 ae 10 b4 63 df 6a 2b  26 cb de 56 10 51 91 79   3···c·j+ &··V·Q·y
0180  38 e8 12 4f 46 71 bf 8d  f1 62 80 64 7c 66 8a 76   8··OFq·· ·b·d|f·v
0190  16 77 82 cb 8c be 86 3e  83 b5 04 98 7a 50 8f bc   ·w·····> ····zP··
```

Transmission Control Protocol: Protocol · Packets: 1275 · Displayed: 45 (3.5%) · Profile: Default

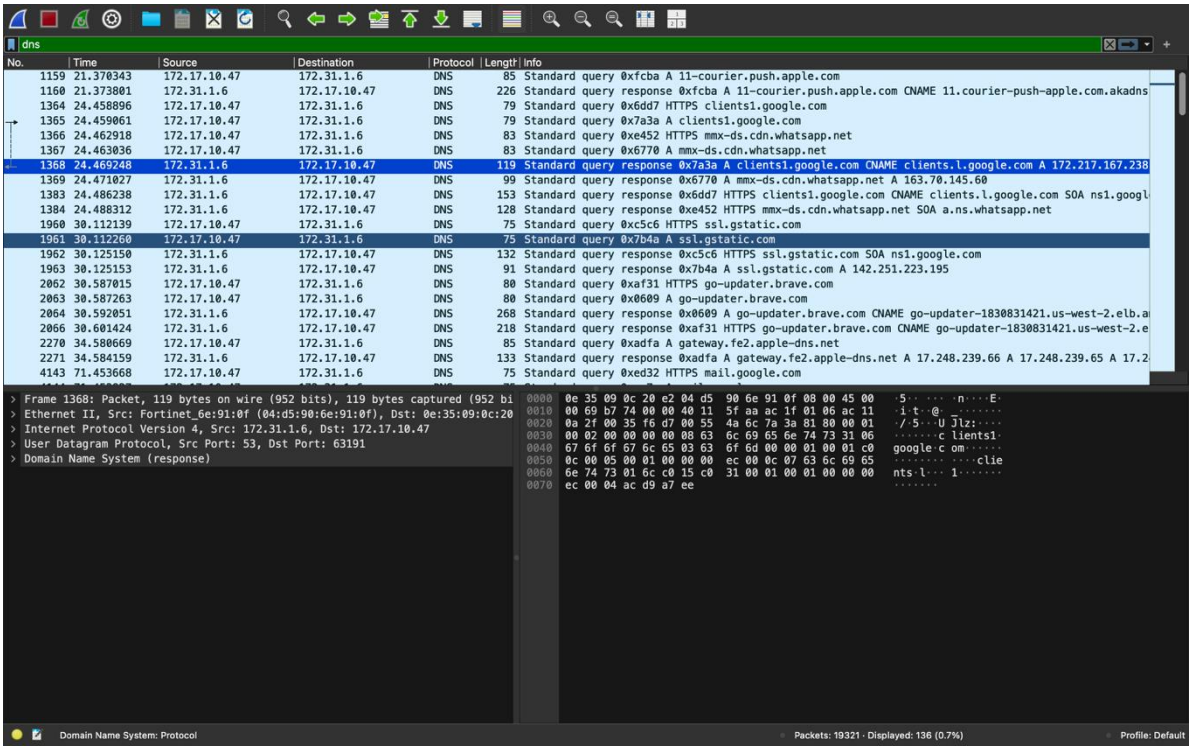
udp

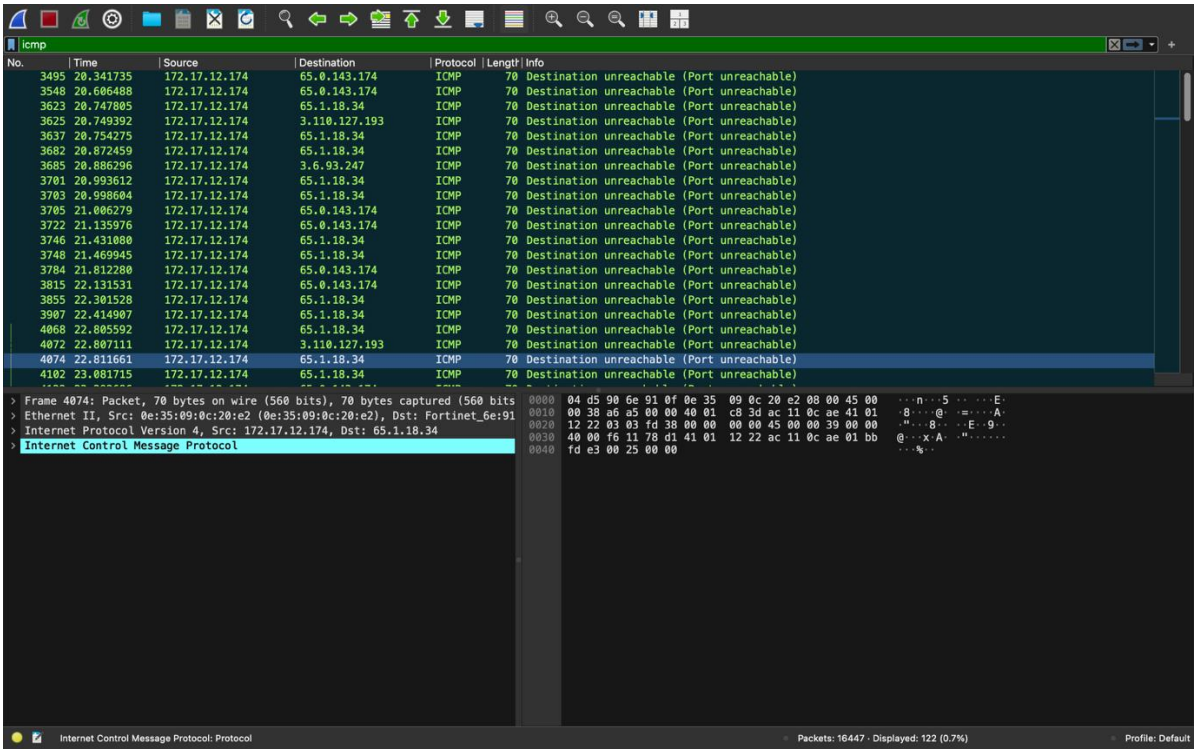
No.	Time	Source	Destination	Protocol	Length	Info
148	1.842387	172.17.11.133	224.0.0.251	MDNS	75	Standard query 0x0000 AAAA DEVCDF52C.local, "QM" question
149	1.842391	fe80::6335:1109:5e…	ff02::fb	MDNS	95	Standard query 0x0000 AAAA DEVCDF52C.local, "QM" question
151	1.851127	74.125.158.105	172.17.12.174	UDP	192	443 → 62305 Len=150
152	1.853402	74.125.158.105	172.17.12.174	UDP	1287	443 → 62305 Len=1245
153	1.853404	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
154	1.853406	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
155	1.853408	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
156	1.853409	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
157	1.853411	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
158	1.853412	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
159	1.853417	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
160	1.855500	172.17.12.174	74.125.158.105	UDP	84	62305 → 443 Len=42
161	1.855602	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
162	1.855604	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
163	1.855605	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
164	1.855607	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
165	1.855608	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
166	1.855610	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
167	1.855611	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
168	1.855613	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250
169	1.856885	74.125.158.105	172.17.12.174	UDP	1292	443 → 62305 Len=1250

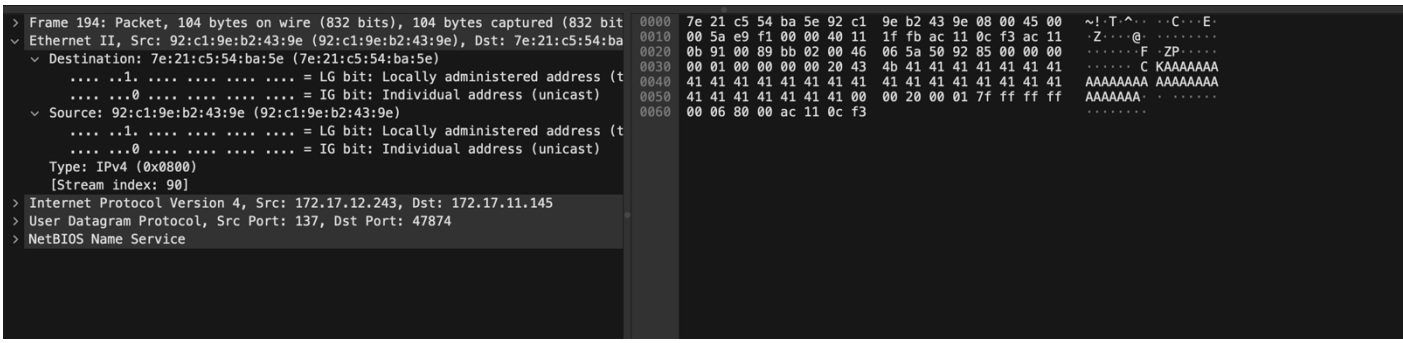
```
> Frame 161: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (103
> Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: 0e:35:09:0c:20
> Internet Protocol Version 4, Src: 74.125.158.105, Dst: 172.17.12.174
> User Datagram Protocol, Src Port: 443, Dst Port: 62305
> Data (1250 bytes)
```

```
0000  0e 35 09 0c 20 e2 04 d5  90 6e 91 0f 08 00 45 00   ·5·· ··· ·n····E·
0010  04 fe 00 00 40 00 3c 11  98 49 4a 7d 9e 69 ac 11   ····@·<· ·IJ}·i··
0020  0c ae 01 bb f3 61 04 ea  55 b3 4d b7 5f 14 37 6f   ·····a·· U·M·_·7o
0030  01 8a 5a 40 56 f6 91 58  10 12 52 f1 c7 9e 40 fc   ··Z@V··X ··R···@·
0040  d6 d1 dd dd 86 4d b6 79  e4 4f a0 ec 16 e2 0a 98   ·····M·y ·O·····
0050  db 33 63 6d 1f 1c f9 2e  cb 5e bb 06 35 08 8f f6   ·3cm···· ·^··5···
0060  b4 6c 2e 69 1b 1f fa 9e  13 76 73 06 b0 ce 7a 2b   ·l.i···· ·vs···z+
0070  d5 da a9 f5 a5 d6 b4 e3  12 1f 5a ae 9b 3b 70 5c   ····· ··Z··;p\
0080  00 8a 42 93 96 8d 72 51  c0 22 5a 5e c7 4b ad 1d   ··B···rQ ·"Z^·K··
0090  6f 06 79 d1 35 b6 0c ed  62 3e e3 ae 99 5e 2c c9   o·y·5··· b>···^,·
00a0  0c d3 77 cf 83 1d bc 64  9a cb 2e 2b 6d 44 83 7e   ··w····d ···+mD·~
00b0  13 f2 19 9c 17 b5 26 91  30 9d 5f 94 6f ca 25 ac   ····· ·&· 0·_·o·%·
00c0  88 aa 11 38 fa 05 99 de  09 1b 19 90 d8 50 d2 c4   ···8···· ····· P··
00d0  e2 f0 39 78 d6 3b aa f9  2d 38 c2 12 da 6f 5a 58   ··9x·;·· -8···oZX
00e0  96 ad 59 5d 02 b2 ba e2  9b c7 b7 20 78 d3 6e f9   ··Y]···· ··· x·n·
00f0  f8 95 3d 2f 1d 27 b7 03  97 3c f3 16 06 4a b7 f5   ··=/·'·· ·<···J··
0100  71 be 2d 2b cf 04 ea e2  c8 43 95 55 5d f7 59 22   q·-+···· ·C·U]·Y"
0110  15 17 24 a3 15 5e 34 2b  a9 f0 b5 9f 69 9f 63 b0   ··$··^4+ ····i·c·
0120  53 41 da 0c 35 2a 86 77  09 40 3a cb d7 28 61 cf   SA··5*·w ·@:··(a·
0130  69 32 35 80 6f 5e 4a 92  8a 6b 4e 2d 2a fb c3 57   i25·o^J· ·kN-*··W
0140  5b 45 ef 0b 92 81 ae cc  f5 14 af 61 86 15 0c fb   [E····· ···a····
0150  97 cc 1f 40 8e bd a0 84  da 3f bd 34 b1 5d 24 54   ···@···· ·?·4·]$T
0160  fd cf 71 15 8f 51 ee 38  8a e5 d2 a7 9e a9 85 58   ··q··Q·8 ·····X
0170  6a d8 a1 3c 31 fd 40 95  5a 04 23 5f 67 4b 75 99   j··<1·@· Z·#_gKu·
0180  f7 21 f9 3a 17 a8 90 0b  07 69 96 b4 8a 2a d1 87   ·!·:···· ·i···*··
0190  96 47 29 91 e0 fe 7c 08  5a 06 b9 15 cc 5f 00 43   ·G)···|· Z····_·C
```

User Datagram Protocol: Protocol · Packets: 4130 · Displayed: 3250 (78.7%) · Profile: Default


dns
No. Time Source Destination Protocol Length Info
1159 21.370343 172.17.10.47 172.31.1.6 DNS 85 Standard query 0xfcba A 11-courier.push.apple.com
1160 21.373801 172.31.1.6 172.17.10.47 DNS 226 Standard query response 0xfcba A 11-courier.push.apple.com CNAME 11.courier-push-apple.com.akadns
1364 24.458896 172.17.10.47 172.31.1.6 DNS 79 Standard query 0x6dd7 HTTPS clients1.google.com
1365 24.459061 172.17.10.47 172.31.1.6 DNS 79 Standard query 0x7a3a A clients1.google.com
1366 24.462918 172.17.10.47 172.31.1.6 DNS 83 Standard query 0xe452 HTTPS mmx-ds.cdn.whatsapp.net
1367 24.463036 172.17.10.47 172.31.1.6 DNS 83 Standard query 0x6770 A mmx-ds.cdn.whatsapp.net
1368 24.469248 172.31.1.6 172.17.10.47 DNS 119 Standard query response 0x7a3a A clients1.google.com CNAME clients.l.google.com A 172.217.167.238
1369 24.471027 172.31.1.6 172.17.10.47 DNS 99 Standard query response 0x6770 A mmx-ds.cdn.whatsapp.net A 163.70.145.60
1383 24.486238 172.31.1.6 172.17.10.47 DNS 153 Standard query response 0x6dd7 HTTPS clients1.google.com CNAME clients.l.google.com SOA ns1.googl
1384 24.488312 172.31.1.6 172.17.10.47 DNS 128 Standard query response 0xe452 HTTPS mmx-ds.cdn.whatsapp.net SOA a.ns.whatsapp.net
1960 30.112139 172.17.10.47 172.31.1.6 DNS 75 Standard query 0xc5c6 HTTPS ssl.gstatic.com
1961 30.112260 172.17.10.47 172.31.1.6 DNS 75 Standard query 0x7b4a A ssl.gstatic.com
1962 30.125150 172.31.1.6 172.17.10.47 DNS 132 Standard query response 0xc5c6 HTTPS ssl.gstatic.com SOA ns1.google.com
1963 30.125153 172.31.1.6 172.17.10.47 DNS 91 Standard query response 0x7b4a A ssl.gstatic.com A 142.251.223.195
2062 30.587015 172.17.10.47 172.31.1.6 DNS 80 Standard query 0xaf31 HTTPS go-updater.brave.com
2063 30.587263 172.17.10.47 172.31.1.6 DNS 80 Standard query 0x0609 A go-updater.brave.com
2064 30.592051 172.31.1.6 172.17.10.47 DNS 268 Standard query response 0x0609 A go-updater.brave.com CNAME go-updater-1830831421.us-west-2.elb.a
2066 30.601424 172.31.1.6 172.17.10.47 DNS 218 Standard query response 0xaf31 HTTPS go-updater.brave.com CNAME go-updater-1830831421.us-west-2.e
2270 34.580669 172.17.10.47 172.31.1.6 DNS 85 Standard query 0xadfa A gateway.fe2.apple-dns.net
2271 34.584159 172.31.1.6 172.17.10.47 DNS 133 Standard query response 0xadfa A gateway.fe2.apple-dns.net A 17.248.239.66 A 17.248.239.65 A 17.2
4143 71.453668 172.17.10.47 172.31.1.6 DNS 75 Standard query 0xed32 HTTPS mail.google.com
Frame 1368: Packet, 119 bytes on wire (952 bits), 119 bytes captured (952 bi
Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: 0e:35:09:0c:20
Internet Protocol Version 4, Src: 172.31.1.6, Dst: 172.17.10.47
User Datagram Protocol, Src Port: 53, Dst Port: 63191
Domain Name System (response)
0000 0e 35 09 0c 20 e2 04 d5 90 6e 91 0f 08 00 45 00
0010 00 69 b7 74 00 00 40 11 5f aa ac 1f 01 06 ac 11
0020 0a 2f 00 35 f6 d7 00 55 4a 6c 7a 3a 81 80 00 01
0030 00 02 00 00 00 00 08 63 6c 69 65 6e 74 73 31 06
0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 01 00 01 c0
0050 0c 00 05 00 01 00 00 00 ec 00 0c 07 63 6c 69 65
0060 6e 74 73 01 6c c0 15 c0 31 00 01 00 01 00 00 00
0070 ec 00 04 ac d9 a7 ee
Domain Name System: Protocol
Packets: 19321 · Displayed: 136 (0.7%)
Profile: Default


icmp
No. Time Source Destination Protocol Length Info
3495 20.341735 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3548 20.606488 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3623 20.747805 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3625 20.749392 172.17.12.174 3.110.127.193 ICMP 70 Destination unreachable (Port unreachable)
3637 20.754275 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3682 20.872459 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3685 20.886296 172.17.12.174 3.6.93.247 ICMP 70 Destination unreachable (Port unreachable)
3701 20.993612 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3703 20.998604 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3705 21.006279 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3722 21.135976 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3746 21.431080 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3748 21.469945 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3784 21.812280 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3815 22.131531 172.17.12.174 65.0.143.174 ICMP 70 Destination unreachable (Port unreachable)
3855 22.301528 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
3907 22.414907 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
4068 22.805592 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
4072 22.807111 172.17.12.174 3.110.127.193 ICMP 70 Destination unreachable (Port unreachable)
4074 22.811661 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
4102 23.081715 172.17.12.174 65.1.18.34 ICMP 70 Destination unreachable (Port unreachable)
Frame 4074: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits
Ethernet II, Src: 0e:35:09:0c:20:e2 (0e:35:09:0c:20:e2), Dst: Fortinet_6e:91
Internet Protocol Version 4, Src: 172.17.12.174, Dst: 65.1.18.34
Internet Control Message Protocol
0000 04 d5 90 6e 91 0f 0e 35 09 0c 20 e2 08 00 45 00
0010 00 38 a6 a5 00 00 40 01 c8 3d ac 11 0c ae 41 01
0020 12 22 03 03 fd 38 00 00 00 00 45 00 00 39 00 00
0030 40 00 f6 11 78 d1 41 01 12 22 ac 11 0c ae 01 bb
0040 fd e3 00 25 00 00
Internet Control Message Protocol: Protocol
Packets: 16447 · Displayed: 122 (0.7%)
Profile: Default

Step 6: Analyze a Packet

- Click on any packet
- Observe:
    - **Ethernet header**
    - **IP header**
    - **TCP/UDP header**
    - Payload data